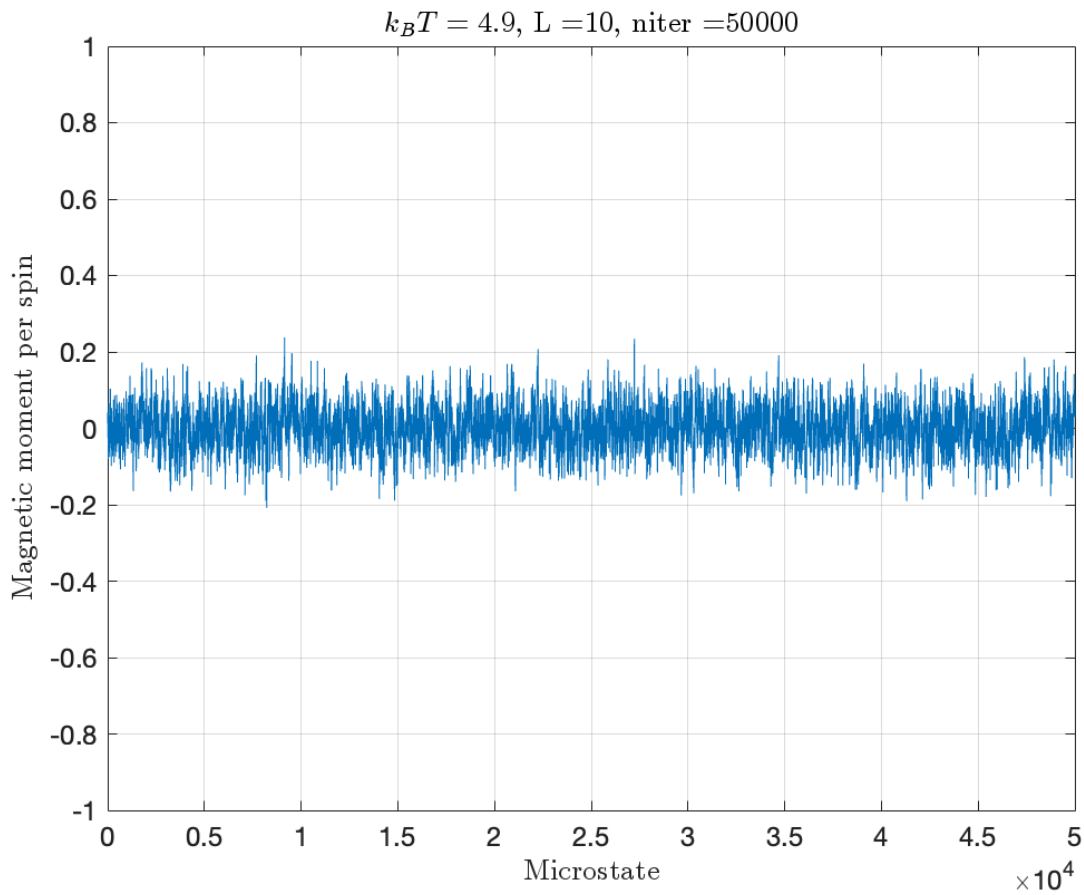
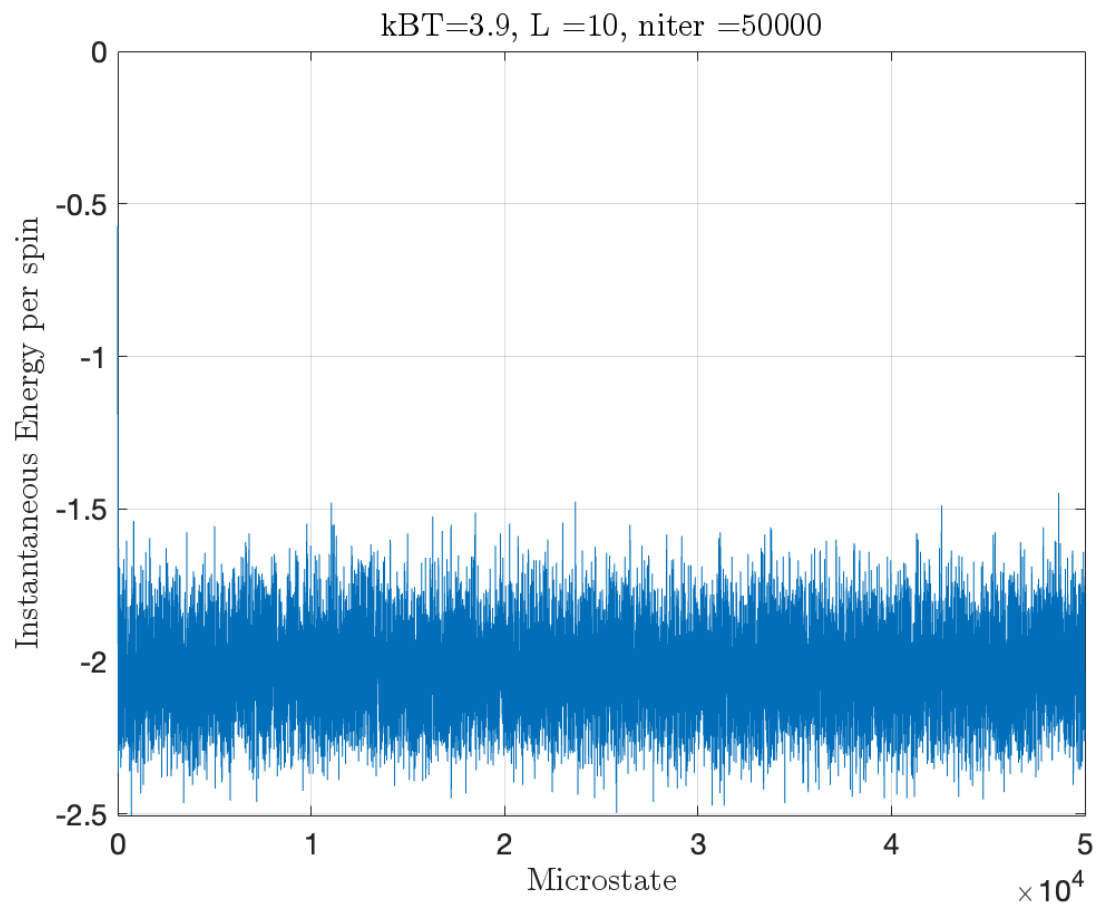


Assignment 3

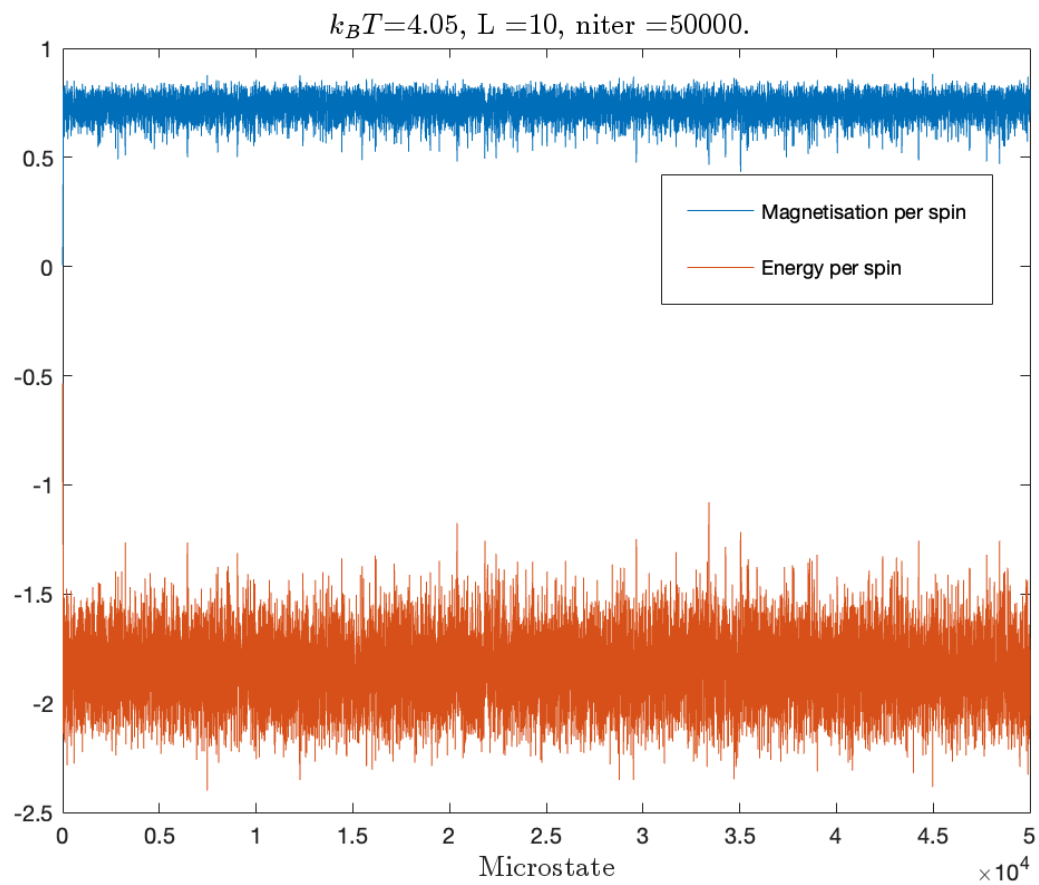
- 1) The total magnetic moment (in simulation units) of the entire lattice in this initial configuration= 8000
- 2)The total energy (in simulation units where $J_{\text{ising}}=1$) of the entire lattice will be = -3000
- 3)The instantaneous magnetization per spin (value of magnetic moment per spin in a microstate: M) fluctuates around the value: $0.008 \approx 0$



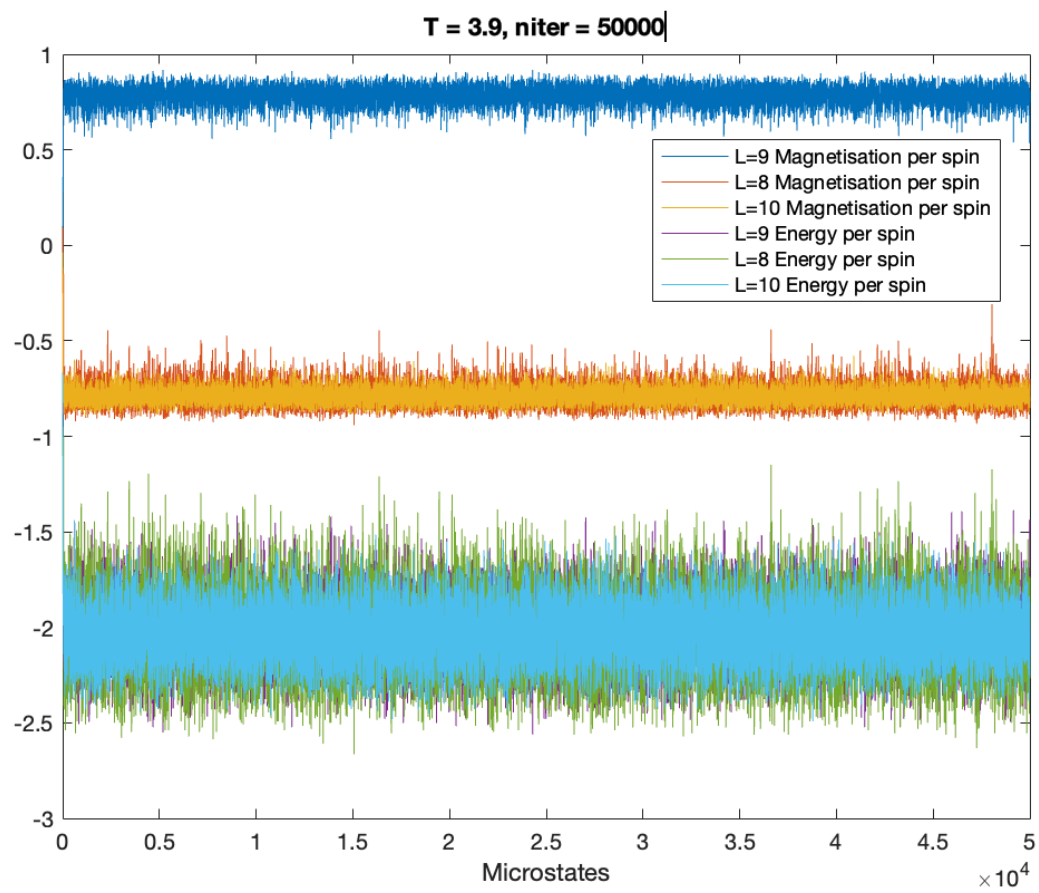
4) The instantaneous energy per spin (value of the energy per spin in a microstate: E) fluctuates around the value: -2.0256

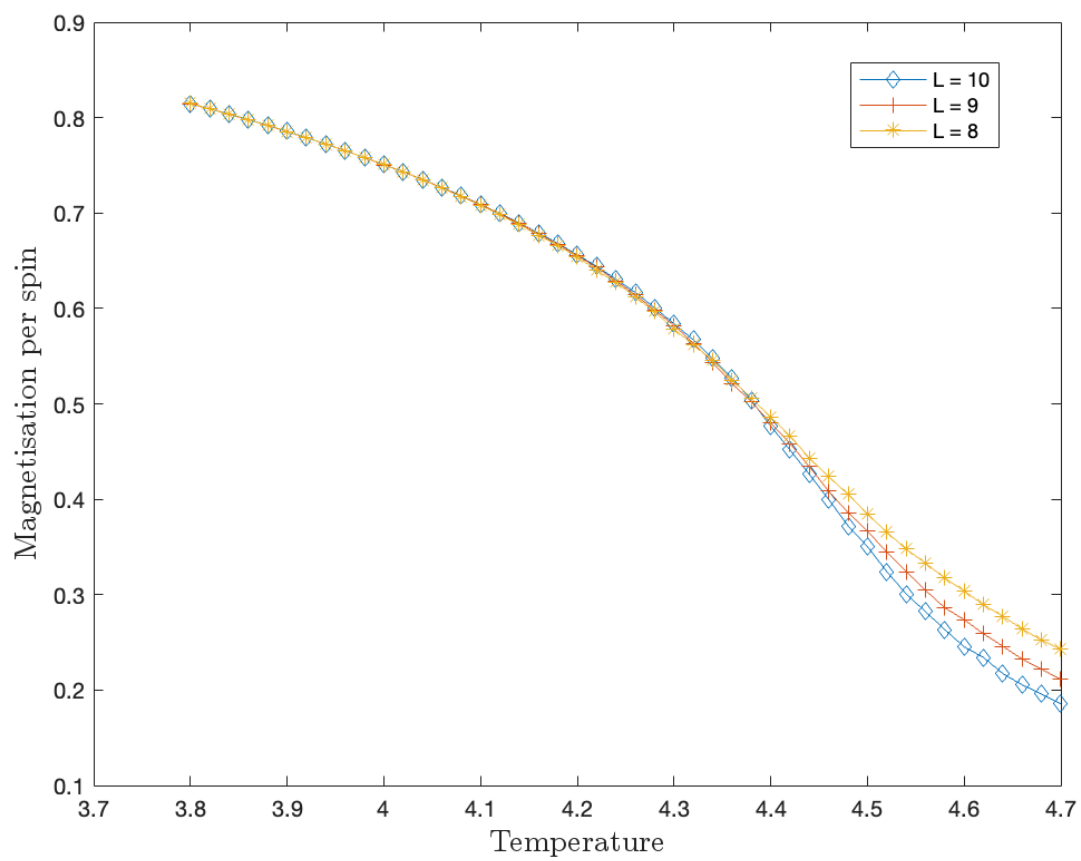


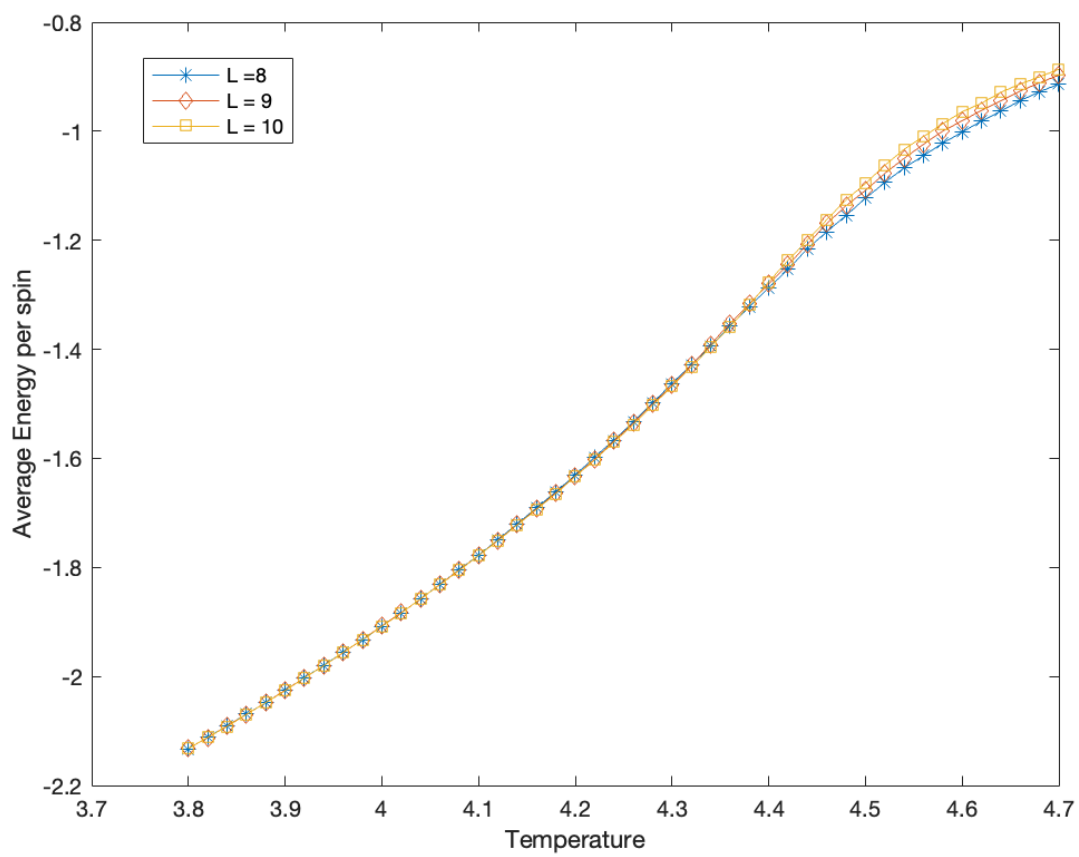
5) Instantaneous Energy E per spin = -1.8452,
Instantaneous Magnetization M per spin = 0.7309

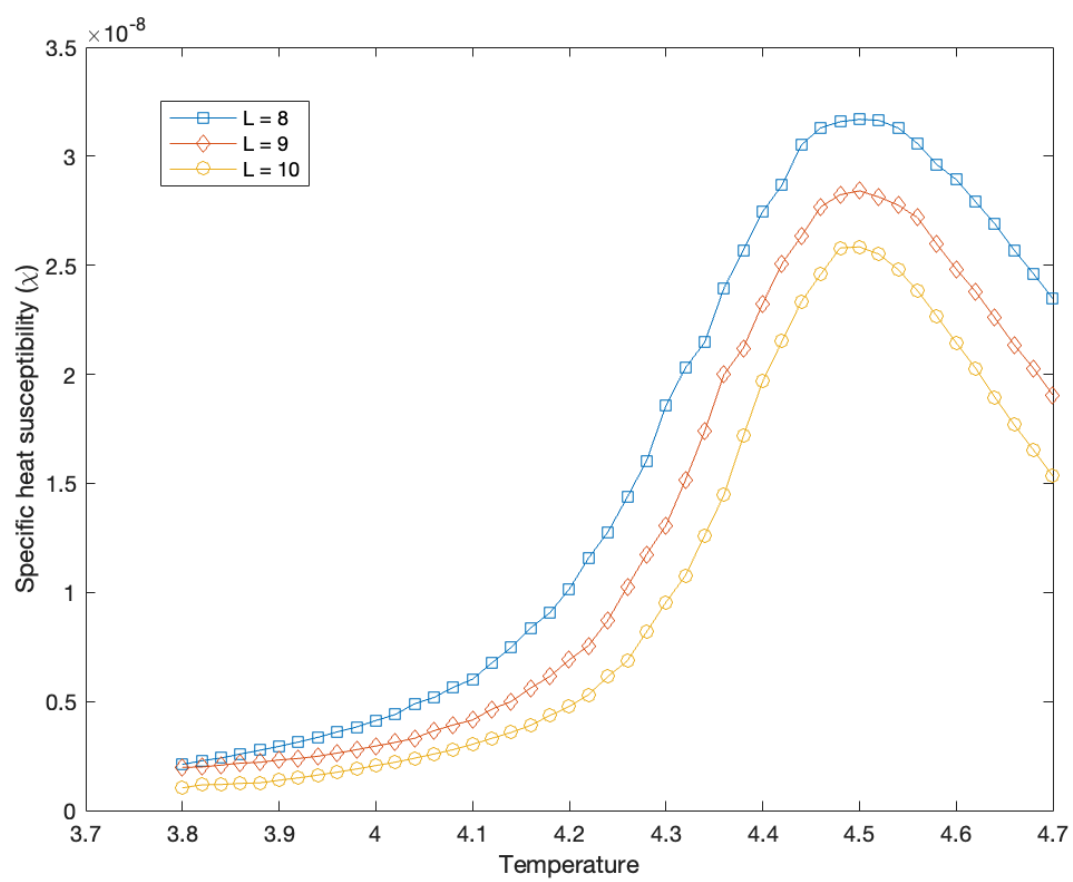


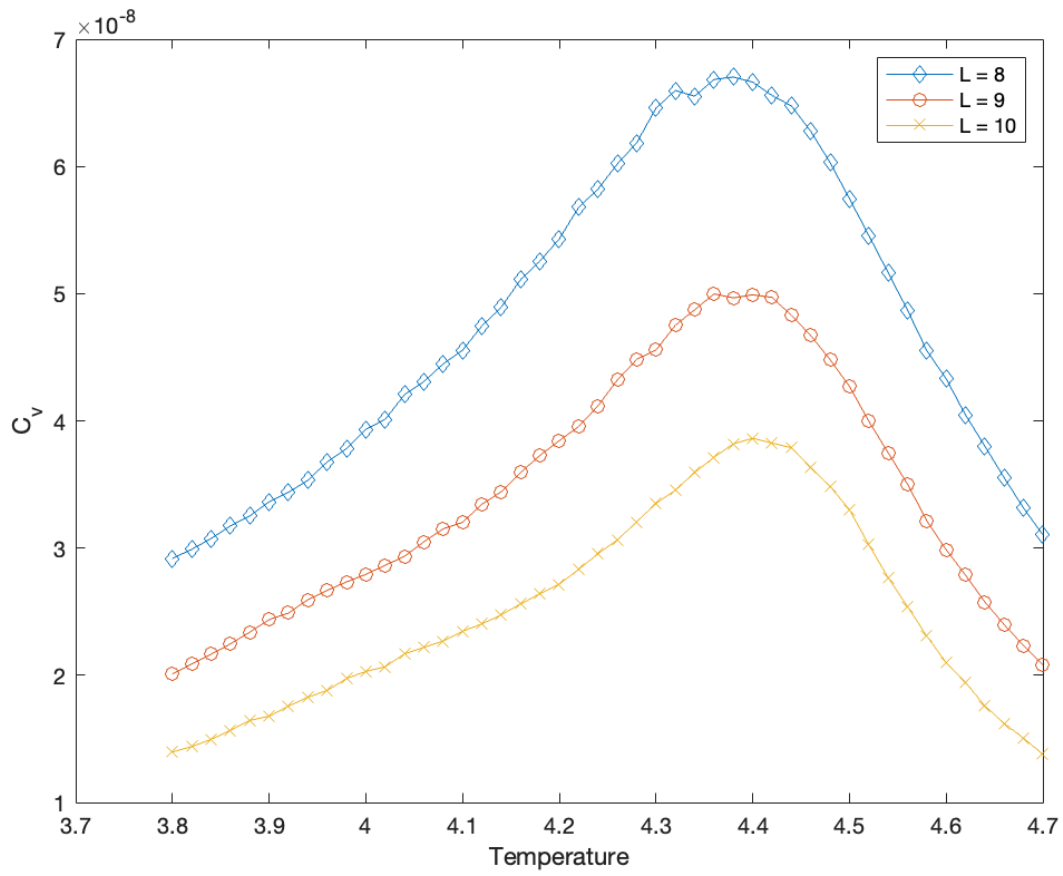
6)











7) $L=8$, $\chi = 3.1689\text{E-}008$

$L=9$, $\chi = 2.8415\text{E-}008$

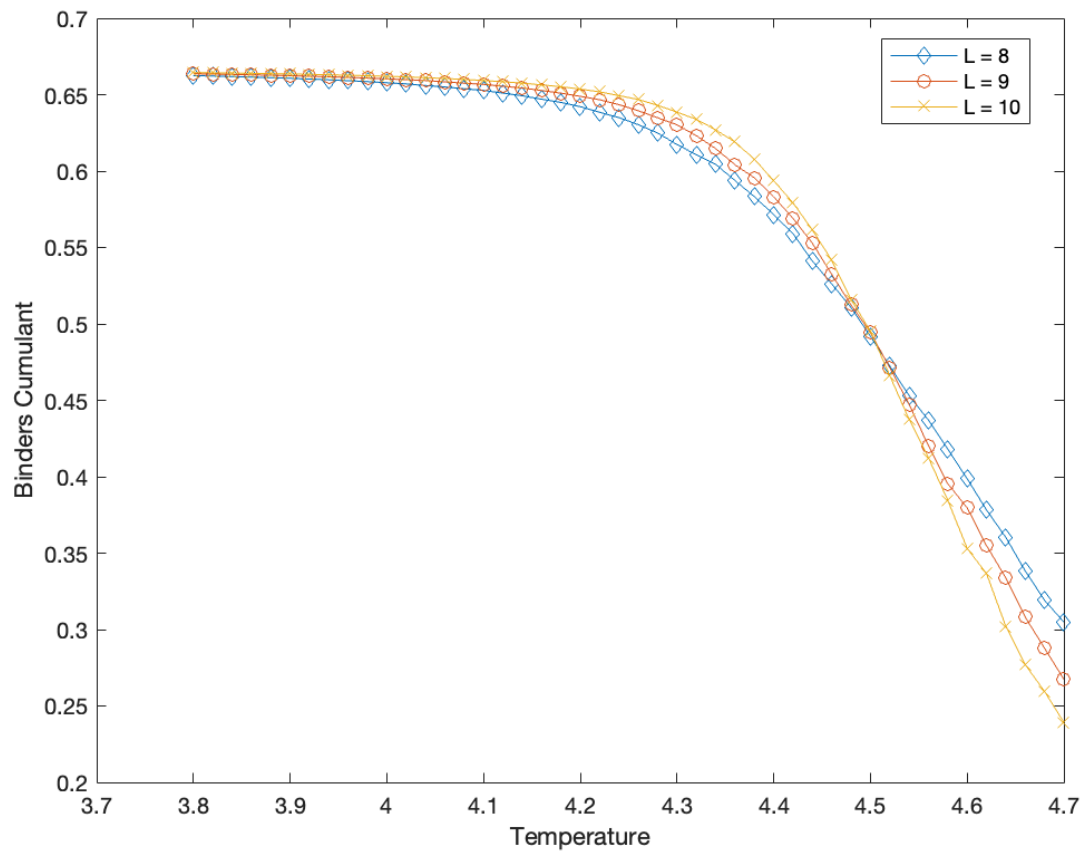
$L=10$, $\chi = 2.5840\text{E-}008$

8) c_v at peak ($L=8$) = $6.7031\text{E-}008$

9) c_v at peak ($L=9$) = $4.9976\text{E-}008$

10) At $T = 3.8$, magnetisation per spin (for $L = 7$) = 0.8150

11)



Q.5) $100 \times 10 / 50 = 20$